Weil positivity on the windows $[-0.8, 0.8]$ and $[-0.9, 0.9]$: an independent open certificate covering both parity sectors

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Abstract

Weil's criterion states that the Riemann hypothesis is equivalent to the positivity of a quadratic form Q on test functions of arbitrary support. Positivity on a fixed window $[-L, L]$ is an unconditional, finite statement that can be settled by a computer-assisted proof. Zhu (2026) certified the window $L = 0.8$; Liu (2026) has since announced certificates for $L = 1$ and $L = 17/16$. We give an independent implementation, with openly available code, that certifies

$$Q(f) \;\geq\; 1.017 \cdot 10^{-17}\,\|f\|_2^2 \quad (L = 0.8), \qquad Q(f) \;\geq\; 2.925 \cdot 10^{-23}\,\|f\|_2^2 \quad (L = 0.9)$$

for all $f \in C_c^\infty((-L, L); \mathbb{C})$. Both parity sectors are certified separately (the odd sector has a negative pole term and a larger margin, $1.13 \cdot 10^{-14}$ and $4.88 \cdot 10^{-20}$). The proof uses Zhu's finite reduction, an even/odd Legendre basis, midpoint–radius ball arithmetic, a Bernstein-ellipse bound for the quadrature error with an explicit digamma estimate, and a Rump-type verification of the smallest eigenvalue. We also record two numerical (uncertified) observations: without the prime terms the form becomes indefinite at half-width ≈ 0.3716, about 7% after the first prime enters at $\frac{1}{2}\log 2$; and the number of negative directions of the archimedean part follows a Slepian count with bandwidth $t_0 = 6.28983\ldots$. Nothing here proves or disproves the Riemann hypothesis.

1 Introduction

For a test function f let $g = f * \tilde{f}$, $\tilde{f}(x) = \overline{f(-x)}$. Weil's explicit formula expresses $\sum_\rho \hat{g}(\gamma_\rho)$ over the non-trivial zeros $\rho = \frac{1}{2} + i\gamma_\rho$ of ζ as a pole term, an archimedean term and a sum over prime powers. Weil [8] showed that the Riemann hypothesis (RH) holds if and only if this quantity is non-negative for all f. Restricting to $\operatorname{supp} f \subseteq [-L, L]$ gives a family of positivity statements indexed by the window L; each is unconditional and involves only the finitely many prime powers $n < e^{2L}$.

For small windows positivity is known by analytic arguments (Yoshida [9]; Connes–Consani [1] at the archimedean place). Zhu [10] reduced the window problem to a finite positive-semidefiniteness check and certified $L = 0.8$ with margin $8.9 \cdot 10^{-18}$; he also observed a doubly exponential decay of the optimal constant in L (a Landau–Widom law), which explains why each further window is costly. Liu [4] announced certificates for $L = 1$ (margin 2^{-151}) and $L = 17/16$.

Contribution. We provide (i) an independent re-implementation of Zhu's reduction that reproduces $L = 0.8$; (ii) a certificate at the intermediate window $L = 0.9$, which cross-checks the two published endpoints; (iii) an explicit treatment of both parity sectors; (iv) fully published code and certificate files. The technical ingredients (Section 3) are standard verified-numerics tools; we state every bound used.

2 Setting and reduction

We use $\hat f(z)=\int_{\mathbb R} f(u)e^{izu}\,du$, Λ the von Mangoldt function, $c_n=2\Lambda(n)/\sqrt{n}$, and

$$A(t)=\operatorname{Re}\psi\big(\tfrac14+\tfrac{it}{2}\big)-\log\pi,\qquad \Psi_L(t)=A(t)-\sum_{\log n<2L}c_n\cos(t\log n),\qquad A_L=\sum_{\log n<2L}c_n.$$

For real f supported in $[-L,L]$ the Weil form reads

$$Q(f)=2\,\hat f(\tfrac{i}{2})\,\hat f(-\tfrac{i}{2})+\frac{1}{\pi}\int_0^\infty \Psi_L(t)\,|\hat f(t)|^2\,dt, \tag{1}$$

and $Q(f)=Q(\operatorname{Re}f)+Q(\operatorname{Im}f)$ for complex f. Under RH, $Q(f)=\sum_\rho|\hat f(\gamma_\rho)|^2$.

Parity sectors. Write $f=f_e+f_o$ (even and odd parts). For real f, $\hat f_e$ is real and $\hat f_o$ purely imaginary on $\mathbb R$, so the integral in (1) has no cross term; and $\hat f_o(-\frac{i}{2})=-\hat f_o(\frac{i}{2})$, so the pole term equals $2\hat f_e(\frac{i}{2})^2-2\hat f_o(\frac{i}{2})^2$. Hence

$$Q(f)=Q(f_e)+Q(f_o),$$

and positivity on the window is equivalent to positivity in each sector. Note that in the odd sector the pole term is *non-positive*.

Lemma 1 (Tail of the archimedean symbol)**.** *For $t\ge1$, $A(t)\ge\log\frac{t}{2\pi}-\frac1t$. Consequently, for $T^\sharp>T_1:=2\pi e^{A_L}$ (and $T^\sharp\ge1$),*

$$\Psi_L(t)\ \ge\ \beta^*:=\log\frac{T^\sharp}{2\pi}-\frac{1}{T^\sharp}-A_L\qquad(t\ge T^\sharp).$$

Proof. Binet's second formula gives, for $\operatorname{Re}w>0$, $\psi(w)=\log w-\frac{1}{2w}-2\int_0^\infty\frac{s\,ds}{(s^2+w^2)(e^{2\pi s}-1)}$. For $w=\frac14+\frac{it}{2}$ we have $|s^2+w^2|\ge|\operatorname{Im}w^2|=t/4$ and $\int_0^\infty s/(e^{2\pi s}-1)\,ds=1/24$, so the integral term is at most $1/(3t)$ in modulus. Moreover $\operatorname{Re}\log w=\log|w|\ge\log(t/2)$ and $\operatorname{Re}\frac{1}{2w}=\frac{1/8}{|w|^2}\le\frac{1}{2t^2}$. Thus $A(t)\ge\log\frac{t}{2\pi}-\frac{1}{2t^2}-\frac{1}{3t}\ge\log\frac{t}{2\pi}-\frac1t$ for $t\ge1$. Since $|\cos|\le1$, $\Psi_L(t)\ge\log\frac{t}{2\pi}-\frac1t-A_L$, which is increasing in t and equals β^* at $T^\sharp$. The condition $T^\sharp>T_1$ means $\log\frac{T^\sharp}{2\pi}>A_L$; positivity of β^* itself is checked in ball arithmetic in every case below. □

Since $\frac1\pi\int_0^\infty|\hat f|^2=\|f\|_2^2$ for real f, Lemma 1 yields Zhu's reduction [10]:

$$Q(f)\ \ge\ R(f):=2\,\hat f(\tfrac{i}{2})\hat f(-\tfrac{i}{2})+\frac{1}{\pi}\int_0^{T^\sharp}\big(\Psi_L(t)-\beta^*\big)|\hat f(t)|^2\,dt+\beta^*\|f\|_2^2. \tag{2}$$

Lemma 2 (Monotonicity)**.** *$t\mapsto\operatorname{Re}\psi(\frac14+\frac{it}{2})$ is non-decreasing on $[0,\infty)$. Hence*

$$\sup_{0\le t\le T}\big|\operatorname{Re}\psi(\tfrac14+\tfrac{it}{2})\big|\le\max\big\{|\psi(\tfrac14)|,\,|\operatorname{Re}\psi(\tfrac14+\tfrac{iT}{2})|\big\}.$$

Proof. From $\psi(z)=-\gamma+\sum_{n\ge0}\big(\frac{1}{n+1}-\frac{1}{n+z}\big)$ we get, for $x>0$,

$$\operatorname{Re}\psi(x+iy)-\psi(x)=\sum_{n\ge0}\frac{y^2}{(n+x)\big((n+x)^2+y^2\big)},$$

which is increasing in y^2. □

3 Method

Basis. Let $\sigma \in \{0,1\}$ denote the sector and $m_k = 2k + \sigma$. The functions

$$e_k(u) = \sqrt{(2m_k+1)/(2L)}\; P_{m_k}(u/L), \qquad u \in [-L, L],$$

form an orthonormal basis of the sector, and

$$\hat{e}_k(t) = c_k\, i^{m_k} j_{m_k}(Lt), \qquad c_k = 2L\sqrt{\tfrac{2m_k+1}{2L}},$$

with j_m the spherical Bessel function. At $z = \pm\frac{i}{2}$: $\hat{e}_k(\frac{i}{2}) = c_k(-1)^{m_k} i_{m_k}(L/2)$ and $\hat{e}_k(-\frac{i}{2}) = (-1)^{m_k}\hat{e}_k(\frac{i}{2})$, with i_m the modified spherical Bessel function. In coordinates $f = \sum a_k e_k$, $R(f) = a^{\mathsf{T}}\mathcal{M}a$ with

$$\mathcal{M}_{kl} = \frac{1}{\pi}\int_0^{T^\sharp} (\Psi_L - \beta^*)\, c_k c_l (-1)^{k+l} j_{m_k}(Lt) j_{m_l}(Lt)\, dt + 2(-1)^\sigma p_k p_l + \beta^* \delta_{kl}, \quad p_k = c_k\, i_{m_k}(L/2).$$

Head block. For $k, l < N$ the integral is evaluated by composite Gauss–Legendre quadrature (P panels, 32 nodes each) with nodes and weights enclosed as Arb balls [2], at 320 or 384 bits. Spherical Bessel values are computed by upward recurrence at the exact ball midpoint with 2048 bits and the argument radius is restored using $|j_m'(x)| \le \frac{1}{2}$ (from $j_m(x) = \frac{1}{2}(-i)^m \int_{-1}^{1} P_m(s) e^{ixs} ds$); whenever the recurrence ball is too wide we fall back to direct Arb evaluation. The quadrature is split into stages; partial sums are stored as (midpoint, radius) pairs with an explicit rounding allowance and summed again in ball arithmetic.

Quadrature error. On each panel of half-length $a = T^\sharp/(2P)$ we use the Bernstein ellipse with semi-minor axis $b = 0.4$ and $\rho = (b + \sqrt{a^2+b^2})/a$; it lies in the disc of radius $r = \sqrt{a^2+b^2} < \frac{1}{2}$ around the panel centre, hence avoids the poles at $t = \pm\frac{i}{2}$ of the analytic continuation $\frac{1}{2}[\psi(\frac{1}{4}+\frac{iz}{2}) + \psi(\frac{1}{4}-\frac{iz}{2})]$ of $\operatorname{Re}\psi(\frac{1}{4}+\frac{it}{2})$. On it, $|j_m(Lz)| \le e^{L|\operatorname{Im} z|}$, $|\cos(z\log n)| \le \cosh(b\log n)$, and $|\psi|$ is bounded via $\psi(z) = \psi(z+1) - \frac{1}{z}$ and, for $\operatorname{Re} w \ge 1$, $|\psi(w)| \le \log|w| + \frac{\pi}{2} + \frac{1}{2|w|} + \frac{1}{9}$ (Binet II with $|s^2+w^2| \ge \frac{3}{4}$). The classical estimate $|I - I_n| \le \frac{64}{15}\, a\, M \rho^{-2n}/(\rho^2-1)$ [7] then bounds every entry, and the Frobenius norm of the error matrix is at most $\varepsilon_{\mathrm{quad}}$.

Smallest eigenvalue of the head block. A candidate λ is obtained from a floating eigensolver at high precision. With $c = 0.99\lambda$ we compute an approximate Cholesky factor $\tilde{L}$ of $\mathrm{mid}(\mathcal{M}_N) - cI$, round it to exact binary numbers and enclose $E = \mathcal{M}_N - \tilde{L}\tilde{L}^{\mathsf{T}} - cI$ in ball arithmetic. Since $\tilde{L}\tilde{L}^{\mathsf{T}} \succeq 0$ exactly, $\lambda_{\min}(\mathcal{M}_N) \ge c - \|E\|_F$ (Rump [5]). A naive interval LDL^{T} fails here because of radius growth.

Tail modes. For $k \ge N$ we use $|j_m(x)| \le x^m/(2m+1)!!$ and $i_m(x) \le x^m e^x/(2m+1)!!$, and Lemma 2 to bound $\sup_{[0,T^\sharp]} |\Psi_L - \beta^*| \le |\psi(\frac{1}{4})| + |\operatorname{Re}\psi(\frac{1}{4} + \frac{iT^\sharp}{2})| + \log\pi + A_L + \beta^*$. This gives ε_B (head–tail coupling), ε_D (tail–tail integral part) and a number π_{tail} with $2\|p_{\mathrm{tail}}\| \le \pi_{\mathrm{tail}}$ (the pole contributes $\pm 2(p\cdot a)^2$, non-negative in the even and non-positive in the odd sector). Writing $f = h + r$ (head and tail),

$$R(f) \;\ge\; \lambda_{\mathrm{head}}\|h\|^2 - 2\varepsilon_B\|h\|\|r\| + \mu\|r\|^2, \qquad \mu = \beta^* - \varepsilon_D - \pi_{\mathrm{tail}}^2,$$

so $R(f) \ge (\min\{\lambda_{\mathrm{head}}, \mu\} - \varepsilon_B)\|f\|^2$. We check $\mu > \lambda_{\mathrm{head}}$ rigorously; the certified constant is $\lambda_{\mathrm{head}} - \varepsilon_B - \varepsilon_{\mathrm{quad}}$.

4 Results

Theorem 3. *For every* $f \in C_c^\infty((-L,L);\mathbb{C})$,

$$Q(f) \geq 1.0174 \cdot 10^{-17}\,\|f\|_2^2 \quad \textit{for } L = 0.8, \qquad Q(f) \geq 2.9256 \cdot 10^{-23}\,\|f\|_2^2 \quad \textit{for } L = 0.9.$$

Proof. By the parity decomposition and (2) it suffices to bound R from below in each sector; the constants are the minima of the certified sector values in Table 1. The proof is computer-assisted; every quantity entering the final inequality is a rigorous ball enclosure. □

	$L = 0.8$		$L = 0.9$	
	even	odd	even	odd
prime powers $n < e^{2L}$	2, 3, 4		2, 3, 4, 5	
A_L; $T_1 = 2\pi e^{A_L}$	2.94197; 119.09		4.38144; 502.3	
$T^\sharp$; β^*	200; 0.513467		550; 0.088724	
modes N (degrees)	200 (0–398)	200 (1–399)	400 (0–798)	400 (1–799)
panels × nodes; precision	400 × 32; 320 bit		1100 × 32; 384 bit	
candidate λ	$1.02769 \cdot 10^{-17}$	$1.14105 \cdot 10^{-14}$	$2.95620 \cdot 10^{-23}$	$4.93421 \cdot 10^{-20}$
Rump residual $\|E\|_F$	$6.1 \cdot 10^{-90}$	$6.1 \cdot 10^{-90}$	$2.0 \cdot 10^{-62}$	$9.5 \cdot 10^{-108}$
$\varepsilon_{\text{quad}}$	$6.2 \cdot 10^{-28}$	$6.2 \cdot 10^{-28}$	$1.0 \cdot 10^{-26}$	$1.0 \cdot 10^{-26}$
ε_B	$1.8 \cdot 10^{-102}$	$3.7 \cdot 10^{-103}$	$4.6 \cdot 10^{-56}$	$1.4 \cdot 10^{-56}$
tail block μ	0.5135	0.5135	0.0887	0.0887
certified	$\mathbf{1.0174 \cdot 10^{-17}}$	$\mathbf{1.1296 \cdot 10^{-14}}$	$\mathbf{2.9256 \cdot 10^{-23}}$	$\mathbf{4.8849 \cdot 10^{-20}}$

Table 1: Certificate parameters and results (files `cert_08_*.json`, `cert_09_*.json`).

Consistency checks. (a) At $L = 0.8$ the even-sector value lies between Zhu's certified lower bound $8.9 \cdot 10^{-18}$ and his upper bound $2.27 \cdot 10^{-17}$; the candidate is stable under $N = 150 \to 200$ and $P = 100, 160, 400$ to 10^{-21}. (b) An independent position-space code (monomial basis on an interval of length $2L$, autocorrelations as exact polynomials, 150 digits; no parity restriction) gives Rayleigh–Ritz *upper* bounds $1.0 \cdot 10^{-14}$, $3.3 \cdot 10^{-18}$, $1.5 \cdot 10^{-20}$ for $N = 24, 32, 40$ at $L = 0.9$, all above the certified value. (c) The fast Bessel routine reproduces the direct Arb evaluation (identical λ at $L = 0.8$). (d) The margin drops by a factor $\approx 3.5 \cdot 10^5$ from $L = 0.8$ to $L = 0.9$; Liu's $2^{-151} \approx 3.5 \cdot 10^{-46}$ at $L = 1$ continues this steep decay, as predicted by Zhu's Landau–Widom picture.

Cost. About 3 minutes per sector at $L = 0.8$ and about 7 minutes per sector at $L = 0.9$ (two cores, Python 3.13, python-flint 0.9.0), plus about 2 minutes for the final verification.

5 Two numerical observations (not certified)

Observation 1 (Archimedean threshold)**.** Drop the prime terms from (1) (keep pole and archimedean parts). Computing $\lambda_{\min}$ in a converged position-space Galerkin scheme (monomial basis, $N = 16$ and 24, 120 digits) and bisecting, the form stays positive up to half-width $L_{\text{arch}} \approx 0.371608$ (window length 0.743216) and is indefinite beyond. The first prime power enters at $\frac{1}{2}\log 2 \approx 0.346574$, i.e. about 7% earlier. Thus, numerically, the archimedean part alone reaches slightly beyond the region where it coincides with the full form, consistent with Connes–Consani [1]. Since Galerkin values are upper bounds for $\lambda_{\min}$, indefiniteness beyond L_{arch} is the reliable direction of this observation; the precise value of L_{arch} is numerical.

Observation 2 (Slepian count of the archimedean deficit)**.** $A(t) < 0$ exactly for $|t| < t_0 = 6.2898359888\ldots$ (close to, but different from, 2π). For window length $\ell = 2L \in [1.5, 3.5]$ the number of negative eigen-directions of the archimedean-plus-pole part equals $\ell t_0/\pi - 1$ (rounded), as expected for a time–band-limiting operator [6, 3]. On these directions the prime terms compensate collectively: several prime powers with contributions of both signs, rather than one prime per direction.

6 Limitations

Theorem 3 is a statement about two fixed windows. It does not prove or disprove RH, and the method does not extend to all L: the margin decays doubly exponentially and the cost grows accordingly. The certificate depends on the correctness of Arb/python-flint and of our code; we therefore publish both, together with the certificate files, and invite independent re-verification.

Data and code. Python sources (`zhu_cert.py`, `zhu_stage.py`, `hp_scan.py`), certificate files and a SHA-256 manifest are available from the author and will be deposited in a public repository (`github.com/sectio-aurea-q`). The manifest is timestamped via OpenTimestamps in the Bitcoin blockchain.

Use of AI tools. Code, computations and the drafting of this manuscript were carried out with the assistance of the AI system Claude (Anthropic). The author is responsible for the content.

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